STUDENT HANDWRITTEN ANSWER SHEET
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Solution for Question 1:

Step 1: Characteristic Equation
    det(A - lambda * I) = 0
    | 4 - lambda   1 |
    | 2   3 - lambda | = 0

Step 2: Determinant Expansion
    (4 - lambda)(3 - lambda) - (1 * 2) = 0
    12 - 4*lambda - 3*lambda + lambda^2 - 2 = 0
    lambda^2 - 7*lambda + 10 = 0   [Characteristic Polynomial]

Step 3: Quadratic Factorization
    (lambda - 5)(lambda - 2) = 0

Step 4: Final Eigenvalues
    lambda_1 = 5,   lambda_2 = 2. [Q.E.D.]